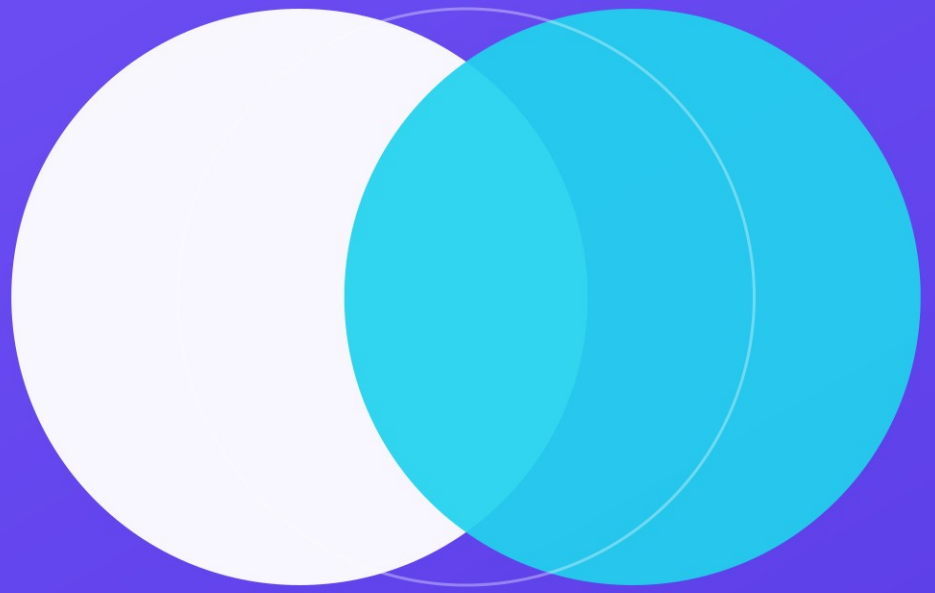
CREATOR AI · LIGHTNING TRACK · ASUS UGEN300

SceneGap Edge

Checks your shot list against your footage, on your own computer, before the crew leaves the location.



- Local-first
- Evidence for every answer
- A person confirms
- UGen300 port planned

Stage I proposal · [Team name] · github.com/[team]/scenegap-edge

PROBLEM

The missing shot is found at 23:00, after the café has closed



Illustrative scenario · two-person crew, café promo, 11 must-have shots

“We had pours. None of them was three seconds long. We found out in the edit.”

09:00	Shot list agreed with the client: storefront, interior, beans, grinder, 3 s pour, two barista takes, bag with readable brand.
17:40	Wrap. Several cards of footage; nobody has time to watch it all on site.
23:00	The editor scrubs for the pour: every take cuts away after 2 s, and nobody filmed a customer drinking.
Next week	Reshoot day: crew, café, client sign-off again.

PROBLEM

Editors can search footage; none checks length, takes or same-frame rules

TOOL	SEARCHES FOOTAGE	FINDS THE MOMENT IN A CLIP	CHECKS DURATION / TAKES / SAME FRAME	RUNS LOCALLY	PRICE
Premiere Pro Media Intelligence	yes	not stated	no	yes	~$22.99/mo
DaVinci Resolve 21 IntelliSearch	yes	visual: whole clips	no	not stated (likely)	Studio $295
Final Cut Pro 12 Visual Search	yes	yes	no	not stated	$299.99
Frame.io semantic search	yes (beta)	yes	no	no (cloud)	$25/member/mo
Jumper	yes	yes	no	yes	from $49/mo
invideo AI script supervisor	matches to shot list	not stated	done/pending only	upload	not listed
SceneGap Edge	**yes**	**yes**	**yes, and says when it can't tell**	**yes**	**see slide 18**

> **Search tells you where a cup appears. SceneGap checks each clip against the shot's rules, for a person to confirm.**

Vendor pages and trade press, accessed 2026-10-26; full list with URLs in docs/research/BUSINESS_FACTS.md.

SOLUTION

From your shot list to a decision you can play back

01

Write the shot list

In Chinese, as the crew speaks. A person confirms the English query for the model and adds rules: min. length, takes, same-frame.

Host CPU

02

Index footage locally

Find camera cuts and sample one frame per second on the original timecode; CLIP turns each frame into a searchable vector.

PC today → UGen300

03

Check each rule

Check length, takes and same-frame rules on the matching frames. When unsure, look again every 0.25 s.

Host CPU

04

Decide with evidence

Candidate / partial / not found / unverifiable, each with playable clips. A person accepts, or adds the shot to the reshoot list.

Host CPU

UGen300 box = planned route A; today the same model runs on the host CPU/GPU and is labelled cpu-/cuda-reference.

SOLUTION

The AI gives one of four answers; only a person can mark a shot done

The AI's answer and the person's answer are stored separately; only the person's counts as done.

Candidate

Every rule passes on at least one segment. Still needs a person to accept.

Partial

Something matched but a rule failed or is undecided: too short, one take of two, a manual check left.

Not found

Nothing reached the threshold in the analysed scope. A reshoot card is proposed, with the nearest frames to check first.

Unverifiable

Files failed to decode or are missing, so absence cannot be claimed. Fix the scope first.

Human: accepted / rejected

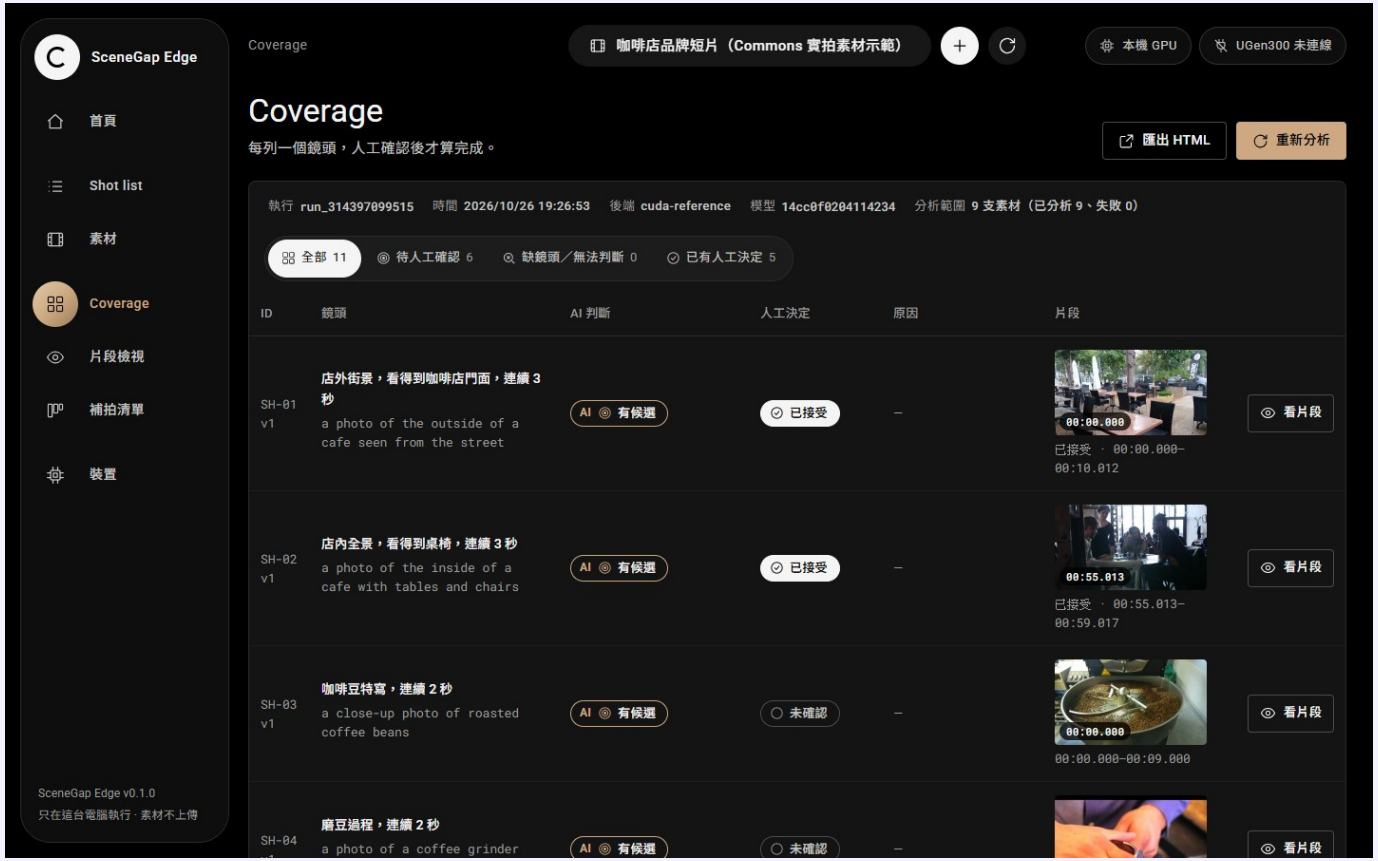
Bound to the requirement revision. Edit the requirement and the old acceptance no longer counts.

Outdated

Requirement, footage, model or threshold changed since the run. The old verdict is hidden, not reused.

SOLUTION

Before leaving, the crew sees which shots are confirmed and which need a look



Live app on 9 Wikimedia Commons clips after a CLIP run and a short human review. Frames are real footage.

Read in one pass

- Each row: the shot, the AI's answer, the person's decision, the clip
- Coverage counts only shots a person confirmed
- Filters: waiting for review, missing, done

SOLUTION

Every answer plays its clip



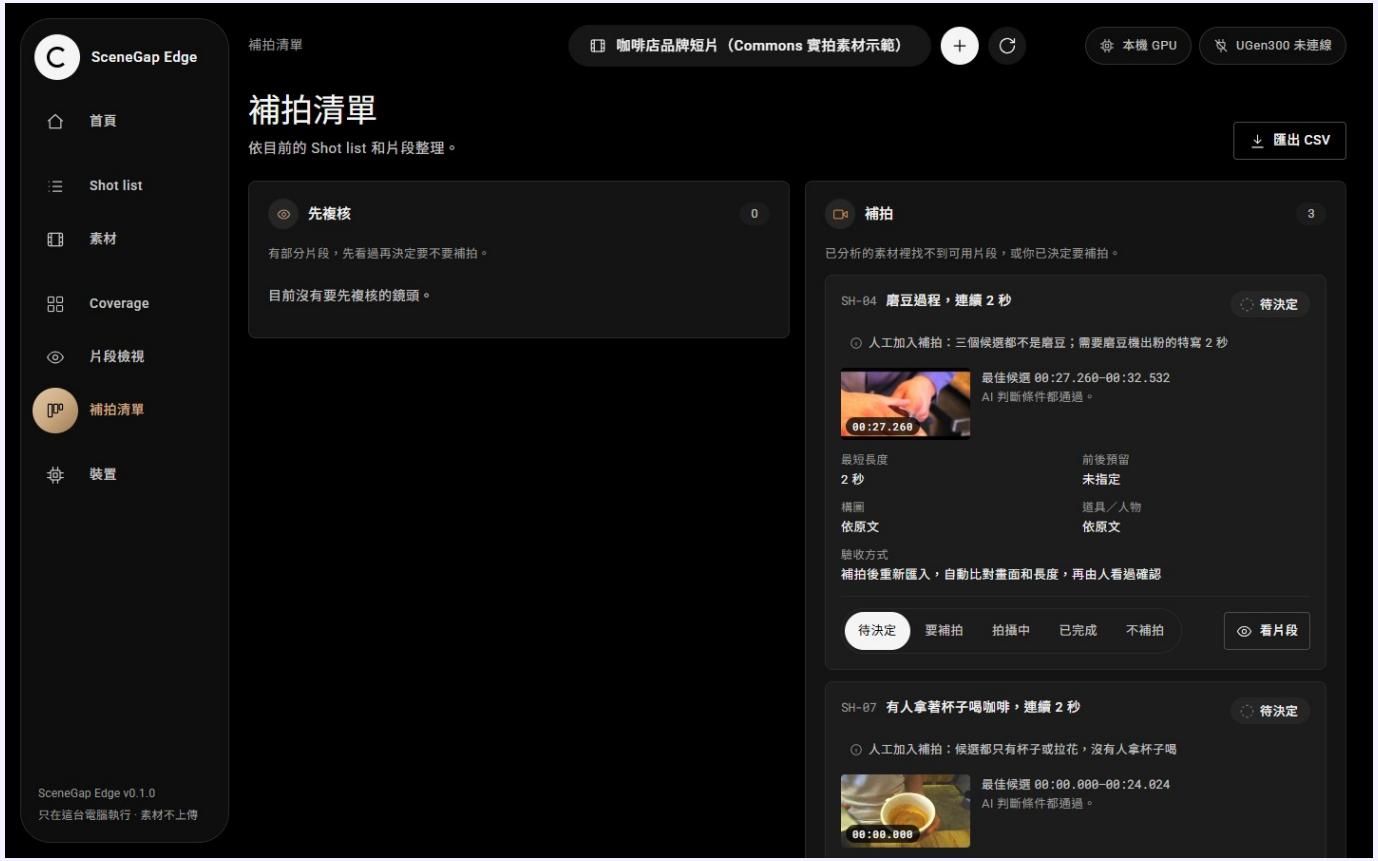
Latte-art clip accepted for 'coffee poured into a cup'; the espresso-only clip before it was rejected (no cup in frame).

One click per decision

- Accept, reject, adjust the range, add to reshoot list
- Filmstrip and clip ranges on the original timecode
- Every decision is logged and can be undone
- Original files are never touched

SOLUTION

Shots nobody could find go on a reshoot list before the crew leaves



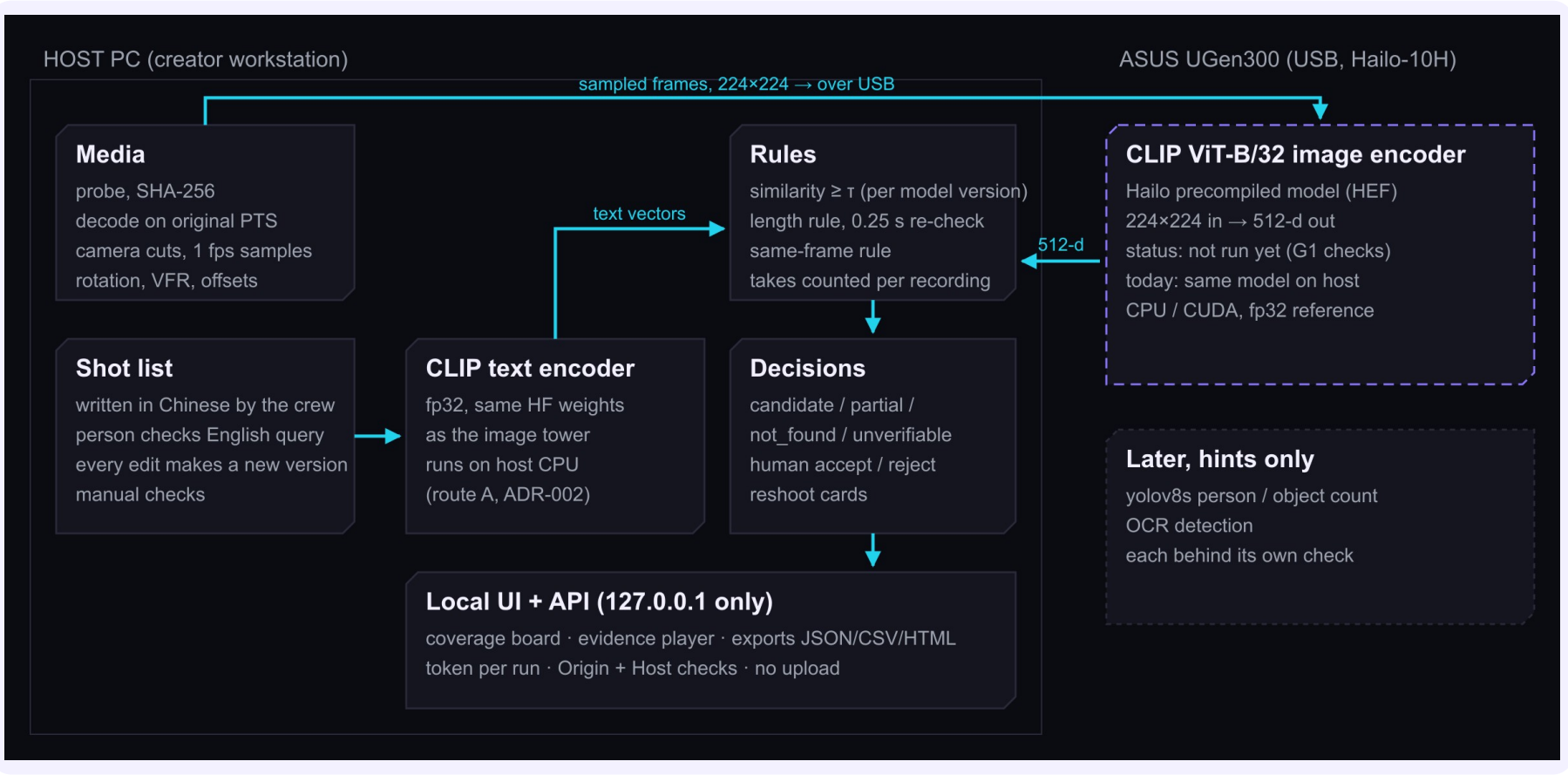
Same demo: a person rejected the grinder and drinking candidates and put them, with the coffee bag, on the reshoot list.

What the crew gets

- The shot, why it is missing, the closest clip
- Minimum length and how it will be checked
- Status: to decide → reshoot → shooting → done

HARDWARE

UGen300 takes the image encoder; the host keeps the rest



Plan (ADR-002): the image half of CLIP ViT-B/32 runs on the UGen300 as Hailo's precompiled model; the text half and every rule check stay on the PC.

HARDWARE

Before we trust the UGen300, its results must match the PC's

G1 = five checks that the device ranks clips the same way as the PC, on the same dev frames.

CHECK	MEASURE (DEV FRAMES, SAME PREPROCESSING)	PASS BAR	STATUS
G1a	Cosine, HEF vs fp32 embedding, per frame	median ≥ 0.98, p5 ≥ 0.95	Not run yet
G1b	Spearman ρ of similarity per query	median ≥ 0.95, worst ≥ 0.85	Not run yet
G1c	Top-5 / top-10 shot overlap	≥ 0.80 / ≥ 0.85	Not run yet
G1d	Dev Recall@5, each backend with its own τ	≥ fp32 − 3 pts	Not run yet
G1f	Frames/s over USB, 60 s clip end to end	reported, no bar	Not run yet
Harness	Same script, CPU vs CUDA fp32 (self-check)	≈ 1.0 expected	cos 1.0000, ρ 0.99999

> **Vendor: Hailo-10H ViT-B/32 image encoder, CIFAR-100 zero-shot 65.3 float → 64.5 on chip (PCIe, not USB).**

docs/MODEL_PLAN.md §2, §4; Hailo Model Zoo (vendor figures).

EVIDENCE

How we tested without fooling ourselves

Step 1

Protocol first

Metrics, split, threshold rule committed to git before any label existed.

Step 2

Shot list

11 café-promo shots and queries written before any frame was seen.

Step 3

Two AI annotators

Independent AI labelling sessions on 1 fps contact sheets; a third pass on every disagreement.

Step 4

Dev only

Threshold τ and crop policy chosen on dev clips, then frozen.

Step 5

Test once

Held-out authors, scored once; the script refuses a second run.

EVIDENCE

Real footage, open licences, not shot by us

44

Wikimedia Commons clips, 25 min, every licence and SHA-256 in the manifest

Licences: CC BY-SA 4.0 ×15, CC BY 3.0 ×9, CC0 ×8, CC BY-SA 3.0 ×4, public domain ×3, CC BY 4.0 ×3, CC BY 2.0 ×2

0.91 agreement between two AI annotators (Cohen's κ; 1 = perfect)

26 / 18 test / dev clips, split by author

11 shots, incl. a same-frame rule and a manual brand check

EVIDENCE

Right clip in the top 5 for 7 of 11 shots; the rule checks cost one hit against plain search

METHOD	RIGHT CLIP IN TOP 5 (RECALL@5)	TOP-5 PICKS THAT ARE RIGHT (PRECISION@5)
Title keyword match	45% (5/11, 95% CI 21–72%)	36% (8/22, 95% CI 20–57%)
Single-frame semantic search	73% (8/11, 95% CI 43–90%)	40% (22/55, 95% CI 28–53%)
SceneGap: + length, re-check, same-frame rules	**64% (7/11, 95% CI 35–85%)**	**33% (18/55, 95% CI 22–46%)**

26 held-out clips, scored once with τ = 0.225. A hit needs 0.5 s of overlap, so Recall@5 does not test the length rules.

eval/results/test_metrics.json; Wilson 95% intervals.

EVIDENCE

Finding missing shots automatically is the open problem

METHOD	MISSING SHOTS CAUGHT	RESHOOT ASKED, SHOT WAS THERE	COULDN'T DECIDE	SAID 'FOUND' WHEN ABSENT
Title keyword match	89%	44%	0%	11%
Single-frame semantic	0%	0%	0%	100%
SceneGap	**0%**	**0%**	**12%**	**79%**

> **The threshold was tuned for single frames, so almost every shot finds some candidate. On dev, a relative score catches 33% of gaps at 11% wrong reshoots; Stage II calibrates per shot.**

eval/results/test_metrics.json; explore_dev_gap_tradeoff.json (dev only).

EVIDENCE

Analysis costs 2.3–3.1 s per minute of video; decoding, not the model, dominates

BACKEND	DEVICE	ANALYSE (DECODE + EMBED)	SEARCH, TYPICAL / SLOWEST 5% (10K CLIPS)	MODEL LOAD
cpu-reference	desktop CPU	3.1 s per minute of video	11.77 / 14.29 ms	6.51 s
cuda-reference	NVIDIA GeForce RTX 4080	2.3 s per minute of video	7.61 / 13.39 ms	1.88 s
UGen300	Hailo-10H	Not run yet	Not run yet	Not run yet

> **Playing a 1080p30 test pattern while analysing: 0 of 1,800 frames late; slowest 5% of frames 5.7 ms idle, 7.5 ms with CPU analysis, 6.2 ms with GPU (budget 33 ms).**

eval/results/perf.json, interference.json.

ENGINEERING

46 automated tests cover media edge cases, logic, security and recovery

30 of the PRD's 36 acceptance cases pass by test; a fresh-environment install passes all 46 tests.

Media edge cases

Rotation, 29.97 fps, VFR, start offsets, corrupt and unsupported codecs, files still copying, moved files found again by hash.

Logic

'Can't tell' stays 'can't tell'; two short clips never count as one long one; duplicates are not two takes; editing a shot voids old decisions.

Security

127.0.0.1 only, per-run token, Origin + Host checks, authorised folders, CSV formula guard, escaped HTML, errors never expose internals.

Offline

Core flow tested with outbound sockets blocked. Footage never leaves the computer.

Failure

Disk full stops safely and resumes; interrupted jobs resume; originals' hashes unchanged.

Rebuild

Fresh venv from the public index passes the suite; Dockerfile and CI included; Apache-2.0.

BUSINESS

Who pays: small crews and freelance editors who shoot on location

SIGNAL	NUMBER	SOURCE
US film and video editors / camera operators (2025)	39,400 / 32,600; 35% self-employed	US BLS OOH
Median pay, editors (May 2025)	$75,420 / yr	US BLS OOH
Day rate, small crew (estimate)	NT$12,000–28,000 (TW); $800–1,500 (US)	PRO360 quotes; rate cards
Creators worldwide	~50M, ~4% professional	Goldman Sachs (estimate, partly verified)
Shot-list practice	crews tick off shots and schedule pickup shots	StudioBinder guide

> **Value per project ≈ review hours saved × rate + avoided reshoot day − setup and review time. Unproven until the Stage II user study.**

docs/research/BUSINESS_FACTS.md (URLs, access dates, verified flags).

BUSINESS

Pricing hypothesis: one-time licence, sold with the accelerator

All prices here are hypotheses to test in interviews (PRD 22.3), not a market claim. ASUS has not endorsed this product.

Perpetual licence

Local-first users dislike per-minute upload pricing. Benchmark: Resolve Studio $295, Final Cut $299.99 one-time.

With UGen300

UGen300 USB reported at $299.99 (CNX Software). A bundle adds a concrete creator workflow to a USB accelerator.

Template packs

Reusable shot-list templates per genre (café, product, event) with tuned queries; yearly updates.

PLAN

Stage II: device, users, Chinese queries

Wk 1–2

G1 on a UGen300

Run the HEF image encoder; pass G1a–d or report the failure and stay on cpu-reference.

Wk 2–3

Speed with the device

Same clips, CPU vs GPU vs NPU; playback interference in a real editing-app timeline.

Wk 3–5

User study

3–5 interviews, then 8–12 crews: manual vs SceneGap, counterbalanced; time, misses, wrong reshoots.

Wk 5–6

Chinese queries

A multilingual model (SigLIP 2), so Chinese shot lists need no English query; adopt only if it finds as many dev shots.

Wk 6+

Hints

Optional person counts and text detection, each behind its own check.

PLAN

What could break, and what we do then

RISK	FALLBACK
HEF fails G1 (quantisation hurts ranking)	Recalibrate with dev frames; TinyCLIP; else ship cpu-reference and say so
False cut splits a long take → false 'too short'	A borderline cut makes the length check say 'can't tell', not 'too short'; the cut is shown
One global τ fits some queries badly	One threshold per model version now; per type of shot next, set on dev
English query drifts from the Chinese requirement	A person checks the English search phrase; multilingual model only after its check
AI annotators, Commons footage, small test split	Wide CIs shown; Stage II adds human labels and self-shot projects
Crews will not write shot lists	Templates; the interview checks this first

Fewer surprises in the edit

SceneGap Edge · checks your shot list against your footage, on your own computer

- Checks the shot list, not just the footage: length, takes, same frame
- Finds the right clip in the top 5 for 7 of 11 shots on held-out footage; a person confirms each one
- Next: the image encoder on UGen300, per-shot thresholds for missing-shot detection, and a user study

APPENDIX

Footage credits (1 of 4)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
000	Cadaques cafe de la habana 2008 august converted YAN3.ogv	Cliché Online (Yan.)	CC BY 3.0
001	Vichy - Grand Café, terrasse sous la grêle.ogv	Romainbehar	CC0
002	Merry X-mas at Kranzler.webm	C.Suthorn	CC BY-SA 4.0
003	Times Cafe inflatable waving character.webm	Andy Li	CC0
004	Rateel cafe music2.webm	Iainsimpsonstewart	CC0
005	Starbucks en la aĉetcentro Kristall (Tjumeno).webm	RG72	CC BY-SA 4.0
006	Interior Paragon Café Katoomba.ogv	Bluedawe	CC BY-SA 3.0
007	Café Lomi, Paris - Vimeo.webm	ideafarm films	CC BY 3.0
008	Operation Ghost Stories - Video 01.webm	FBI – Federal Bureau of Investigation	Public domain
009	Coffee Bean & Tea Leaf Process.webm	Kelliks	CC BY-SA 3.0
010	“走线客”过境墨西哥 不忘来杯星巴克 .webm	美国之音中文网	Public domain

APPENDIX

Footage credits (2 of 4)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
011	Color sorter coffeebeans.webm	AlexanderKlink	CC0
012	Kaffe - Ystad-2024.ogg	Jonn Leffmann	CC BY 3.0
013	Coffee from dan 2.webm	Denis kasozi	CC BY-SA 4.0
014	Traditional Ethiopian Coffee Roasting Ceremony.webm	Keria T	CC BY-SA 4.0
015	Despulpadora.ogg	Jhonespe150102	CC BY-SA 4.0
016	Lions Almelo zamelt koffie in voor Voedselbank.webm	AAVISIE	CC BY 3.0
017	Sunday Morning Espresso Routine - Grind + distribute + tamp + extraction (15-Second Abridged Version).webm	Scott Schiller from San Francisco, CA, USA	CC BY 2.0
018	Кофемельница JURA 43.webm	Gontcharik.petr	CC BY-SA 4.0
019	MoliendoCafeEnLicuadora ohs02.webm	PetrohsW	CC BY-SA 4.0
020	Espresso video.ogv	Rosatrieu	CC BY-SA 3.0
021	Macro Espresso Process.webm	Scott Schiller from San Francisco, CA, USA	CC BY 2.0

APPENDIX

Footage credits (3 of 4)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
022	Der perfekte Espresso aus dem Siebträger.ogv	Max Duchardt, Katrin Engel	CC BY 3.0
023	Pour Over.ogv	Lev Lazinskiy	CC BY-SA 4.0
024	Use of a coffee press.ogv	Daniel Case	CC BY-SA 3.0
025	Schritt für Schritt - Kaffeezubereitung mit der Chemex.ogv	Max Duchardt, Katrin Engel	CC BY 3.0
026	Latte art flower - 01.ogv	Anna Frodesiak	CC0
027	Latte art leaf - 01.ogv	Anna Frodesiak	CC0
028	Nespresso machine.ogv	Kjetil Ree	CC0
029	A man pours Arabic coffee.ogv	Sarah Canbel	CC BY-SA 4.0
030	Servido tradicional de café etíope.webm	ProtoplasmaKid	CC BY 4.0
031	How to brew Coffee using the laws of physics - with the Hario Syphon Technica.webm	Coffee Circle	CC BY 3.0
032	Sugar falling into coffee.webm	Anselm Schüler	CC BY-SA 4.0

APPENDIX

Footage credits (4 of 4)

#	WIKIMEDIA COMMONS FILE	AUTHOR	LICENCE
033	Bosnian coffee with rahat lokum, Kujundziluk and Old Bridge in Mostar - Bosnia and Herzegovina (2026)-bs.webm	Minja Klarić	CC BY-SA 4.0
034	A cup of Kenyan Coffee.webm	Queen Asali	CC BY-SA 4.0
035	Latte salute.webm	The White House	Public domain
036	Turkish tea 2015.webm	Klapi	CC BY-SA 4.0
037	Jane pouring beer fast.webm	Angulidayaaluta	CC BY-SA 4.0
038	Turkovich Family Wines - June 2019 - Stierch 03.ogv	Missvain	CC BY 4.0
039	Drinking glass symphony.ogv	IMG 4512	CC0
040	Thermochromic mug.webm	The wub	CC BY-SA 4.0
041	Empty Streets of Paris COVID19.webm	CreatorsCloud	CC BY 3.0
042	IrishTraditionalMusic-IrishPub-Tokyo-2016-2-24.webm	Nesnad	CC BY 4.0
043	Couple holding hands.webm	YouTube user “Editor”	CC BY 3.0